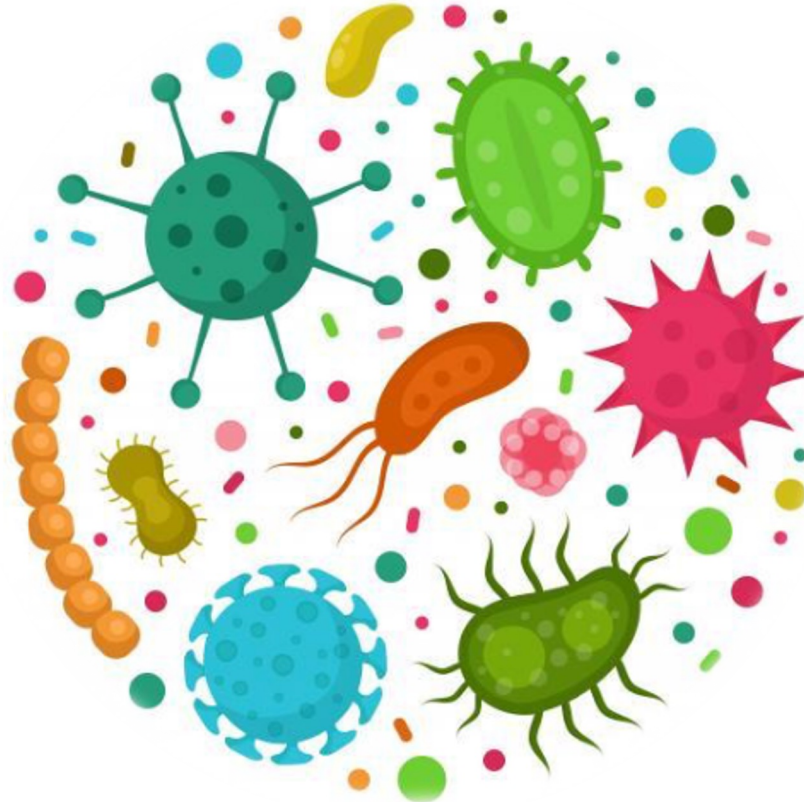


α -diversity

j3

– 30.09.26 –

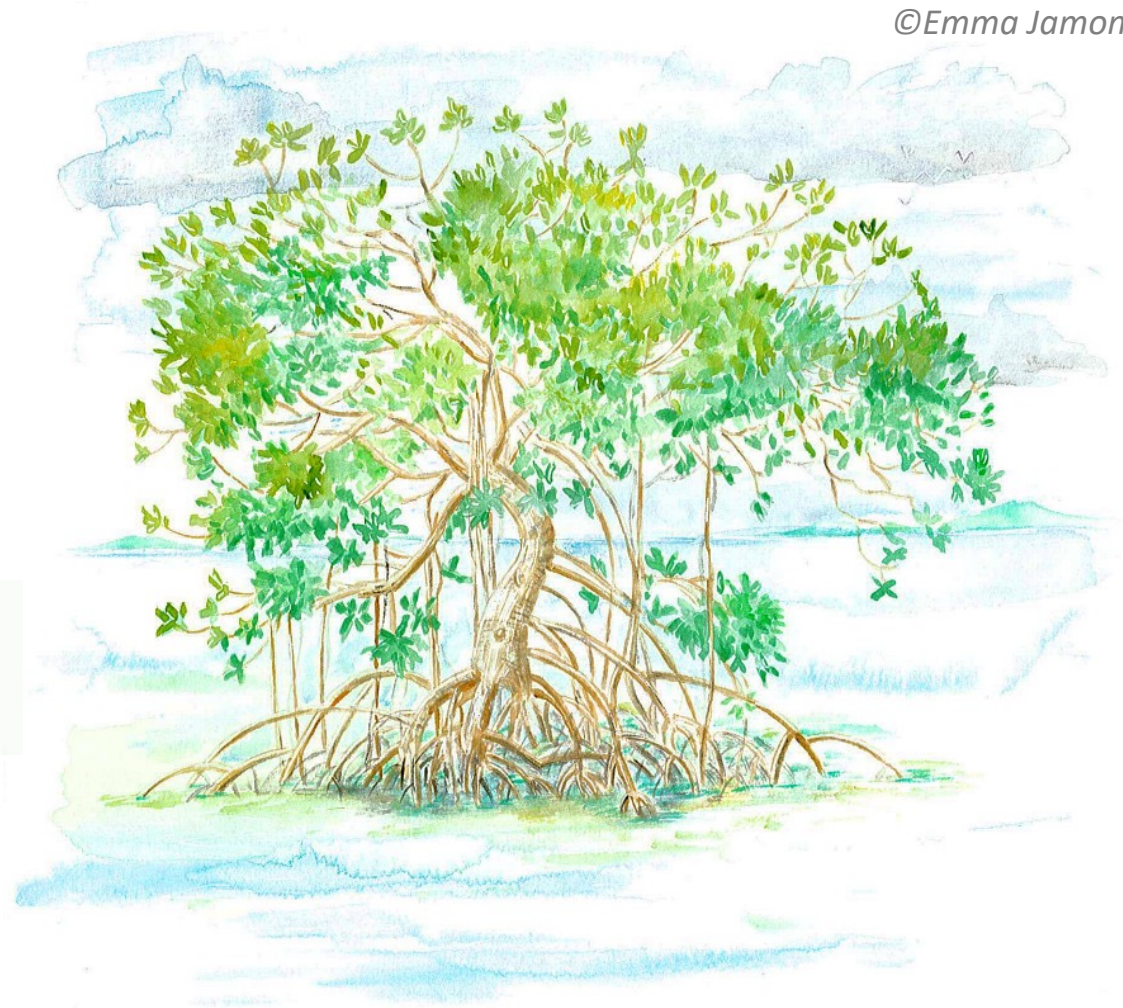


Plateforme OMICS - MIO

Bio-informatique & Sciences de l'Environnement : Exploration de la Diversité Taxonomique des Ecosystèmes par Metabarcoding

What is Biodiversity?

The **variety** and **variability** among living organisms on site, ecosystem and their **interactions** between them



Diversity can be use to describe **variation** in several forms:

- **Genetic**
- **Taxonomic**
- **Functional** group (e.g. nitrogen-fixing)

Why measuring Biodiversity?

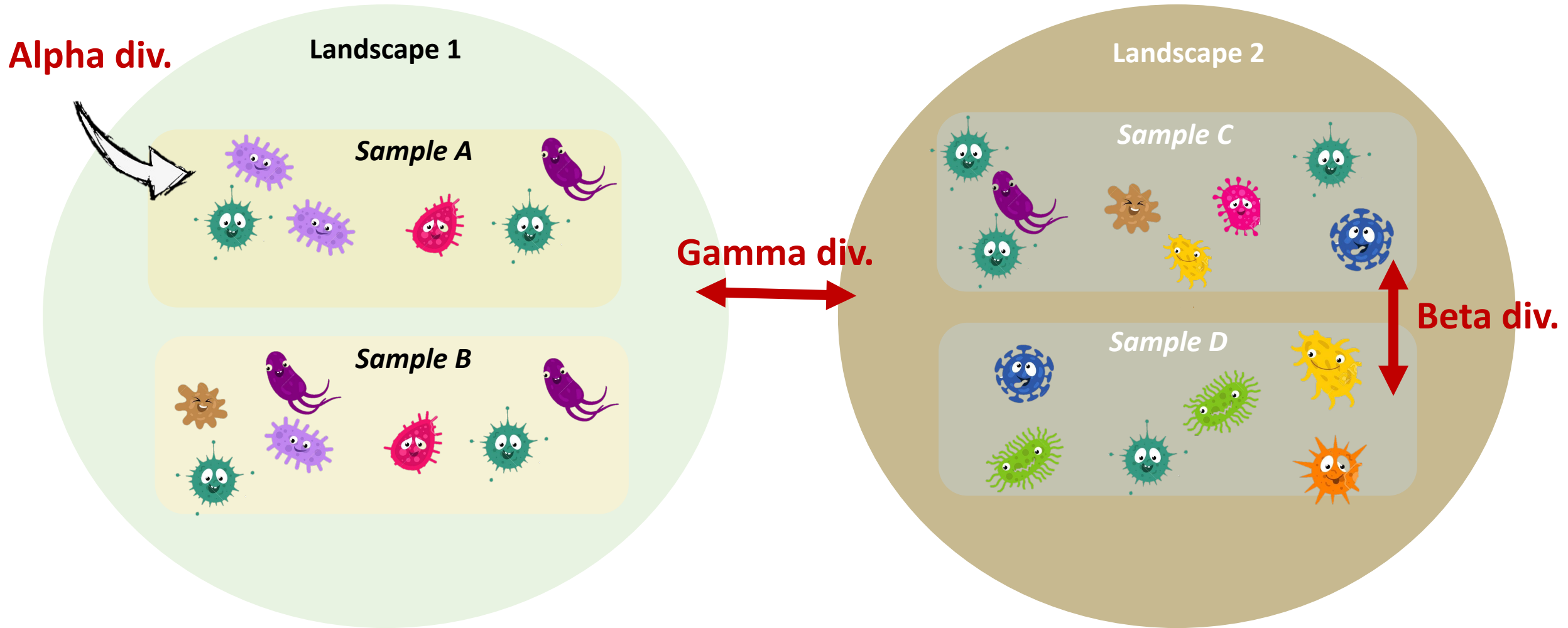
Biodiversity ^{proxy}  **Ecosystem health**

In general, diverse communities are believed to have increased stability, increased productivity, and resistance to invasion and other disturbances



Alpha vs Beta vs Gamma Diversity (Whittaker, 1972)

Alpha, beta and gamma diversity are three types of biodiversity measures described over a special scale



α -Biodiversity ?

α -diversity is local diversity, measured within a closed system

→ **The diversity within an habitat of fixed size**

α -biodiversity has **two components**

- **Richness**
- **Evenness**

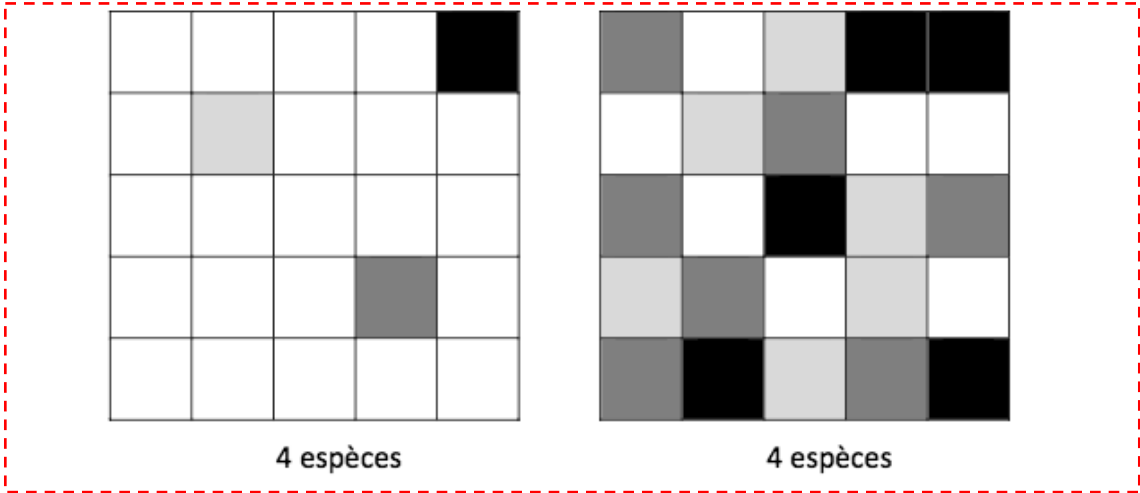
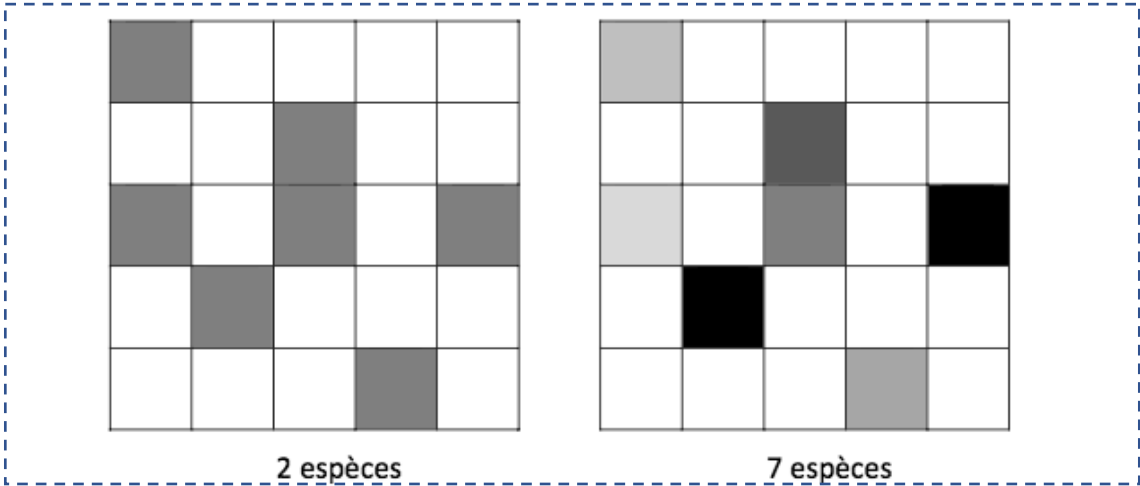
And another one...

- **Disparity (A. Stirling)**

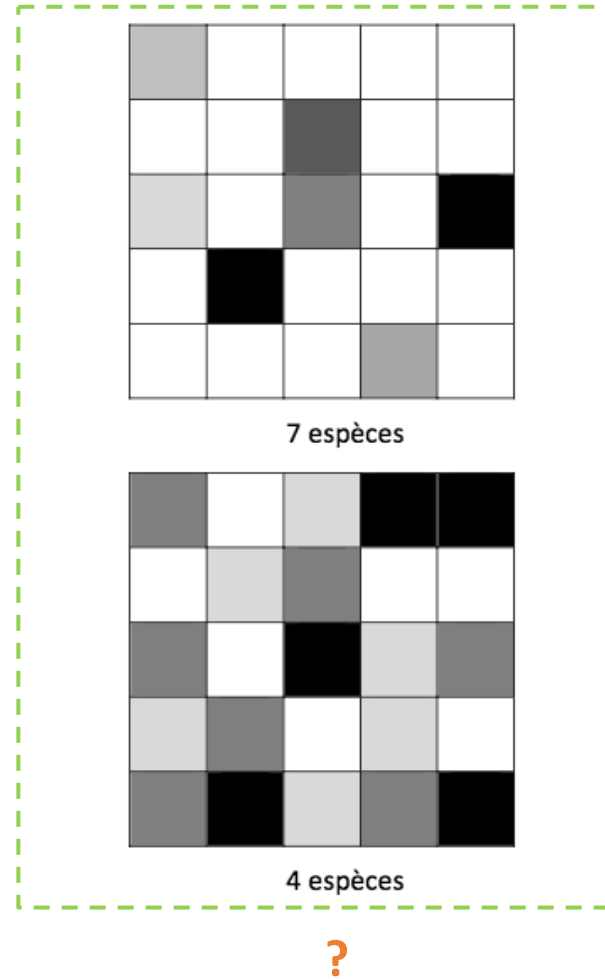
→ **Follow the evolution of populations over time, but also to compare them between the stations studied**

Components of Diversity

Species richness



Species evenness



Two components are necessary for describing diversity

- **Richness**
- **Evenness**

Definition

Specific Richness (S) = The measurement of the **number of species** present in a location/studied system

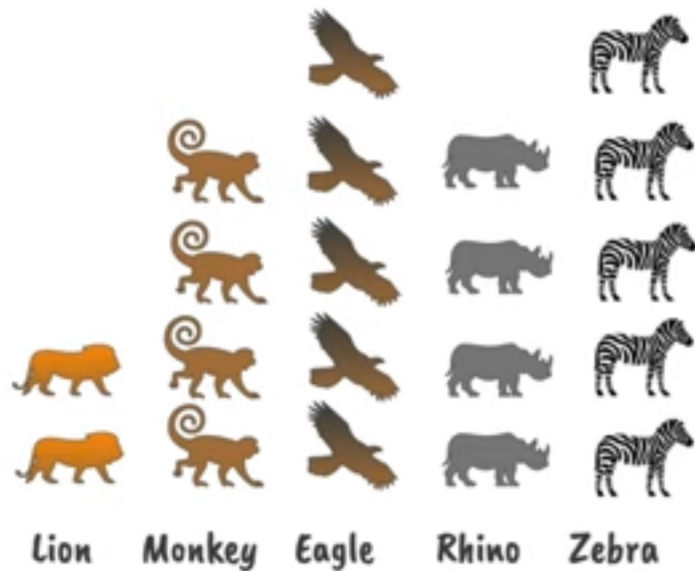
- The more species present, the richer sample is
- Gives equal weight to species which have many/few individuals

Evenness (equitability) = **Relative population of each species**

- Species represented by many individuals or by few ones do not give the same contribution

Evenness index is independent of Richness!!

Higher biodiversity



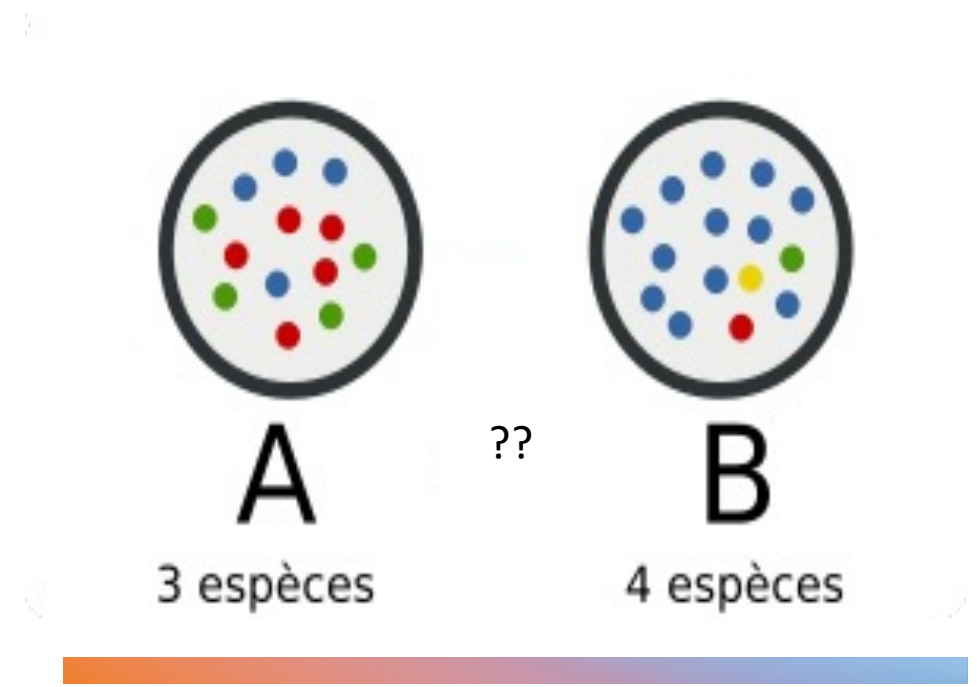
Wide range of species = High richness
Relatively **equal** proportion = high evenness

Lower biodiversity



Few species = low richness
Unequal proportion = low evenness

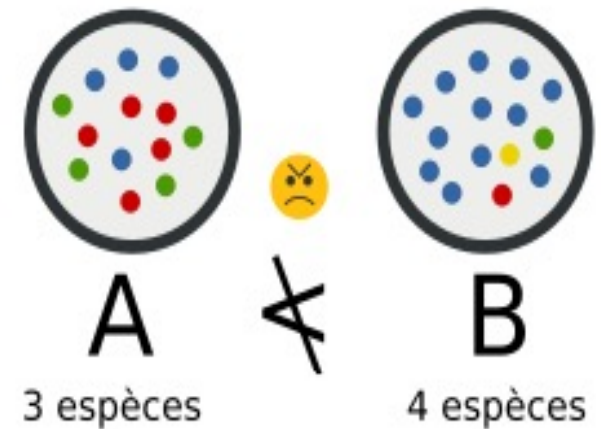
Which one is more diversified ?



Difficult to judge by only looking the numbers !!



Use α -diversity indices to quantify the biodiversity of an habitat/sample



Neutral Diversity Indices (=Taxonomic)

Why neutral?

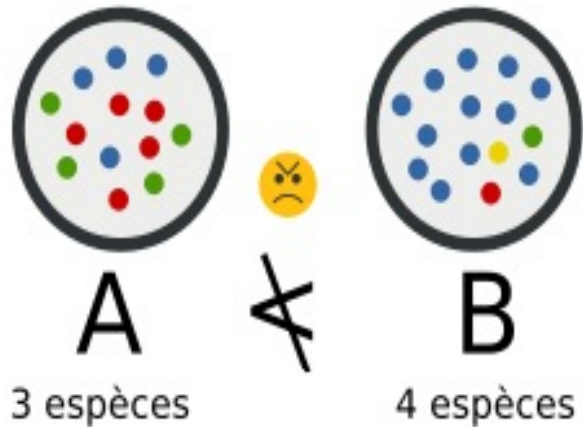
- Species are considered as equidistant (= do not consider species relatedness)

Common indices are Shannon, Simpson and specific richness

- **Subjected to estimation bias because of species not sampling**
- Suppose that the delineation of « species » is clear!
→ It's not!
 - Species concept definition (phylogenetic, biological, ecological niche)
 - Methodological issues

Shannon-Weaver Index: Combine Richness & Evenness

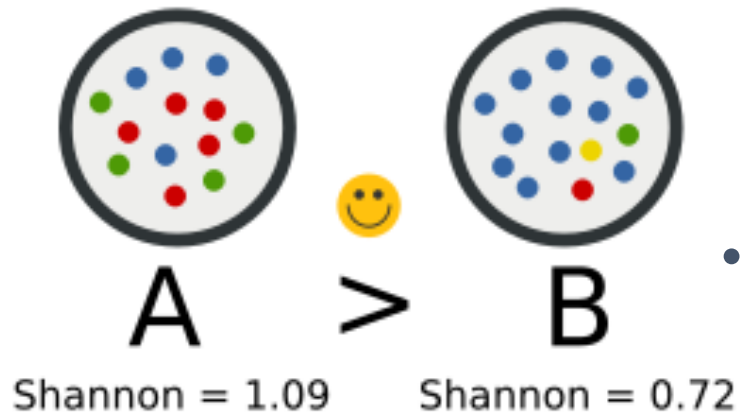
$$H(X) = H_2(X) = - \sum_{i=1}^n P_i \log_2 P_i. \quad P_i : \text{relative abundance of the species } i$$



- A: consists of 3 species, of which 4 green, 5 red & 4 blue
The Shannon index will be :

$$-\left(\frac{4}{13} \log\left(\frac{4}{13}\right) + \frac{5}{13} \log\left(\frac{5}{13}\right) + \frac{4}{13} \log\left(\frac{4}{13}\right)\right) = 1.09$$

Finally, after estimating Shannon for B sample ...



- B consist of 4 species, of which 1 green, 1 red, 1 yellow & 11 blue

Shannon value: Influenced by richness

H is generally between **1.5 - 4**

H gets higher when :

- There are **more species**
- There is **greater evenness**

There's no upper limit to the index

The maximum : All species have the same number of individuals (perfect evenness)

$H_{\max} = \log(S)$

100 species : $\log(100) = 4.6$

1000 species : $\log(1000) = 6.9$

Equitability Pielou Index... with Shannon Scores

Shannon is **dependent on species richness!**

→ Pielou index is independent of species richness

$$\text{Pielou index} = \frac{\text{Shannon Index } (H)}{\log(S)}$$

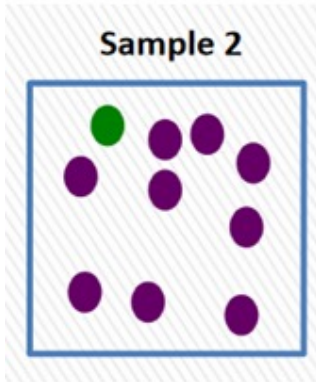
S is Species Richness

Pielou index is a way to measure how the species are evenly distributed in a community

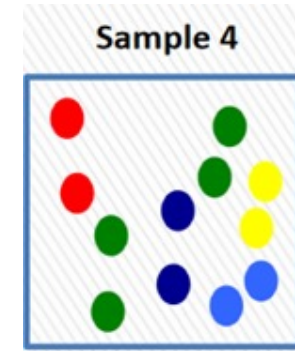
Range 0 to 1 (max equitability, habitat diversity increase)

Simpson's Index of Diversity

Idea : Indicates the taxa dominance and gives the probability of two individuals that belong to the same taxa being randomly chosen



A value of 0.8 ...
2 sequences randomly selected
have 80% chance to belong to the same ASV!



Simpson index = D

$$D = \sum_{i=1}^S p_i^2$$

P = proportion of the species

Gini-Simpson = 1 - D

$$E = 1 - \sum_{i=1}^S p_i^2$$

- Influenced by highly abundant Taxa
- Greater weight on evenness
- Range 0 to 1 (high)

0

No diversity

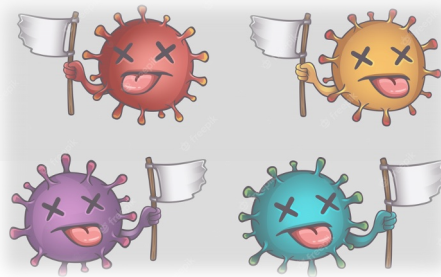
The closer the value to 1, the more diverse the habitat is

1

Infinite Diversity

Low sp. Diversity :

- Few successful species in the habitat
- Env stressful, few niches, few organisms well adapted to env
- Any change in env may have serious effect on ecosystem



High sp. Diversity :

- Great number of successful species, more stable ecosystem
- Env is less likely to be hostile
- Complex food
- Env change is less likely to damage the whole ecosystem



Diversity Estimators

- Chao1 & ACE are non-parametric estimators of taxa richness
- Sampling at infinity
- Good sampling gives you a total number of ASV/OTU observed not far from the Chao1 / ACE value (predicted for the sampled environment)

$$\text{Chao1} = S_{\text{obs}} + \text{Adjustment (linked to the rare)}$$



Chao1 adjustment

Idea : Rare taxa bring most information about the number of missing taxa

$$\frac{F_1(F_1 - 1)}{2(F_2 - 1)}$$

Singletons = F_1
Doubletons = F_2

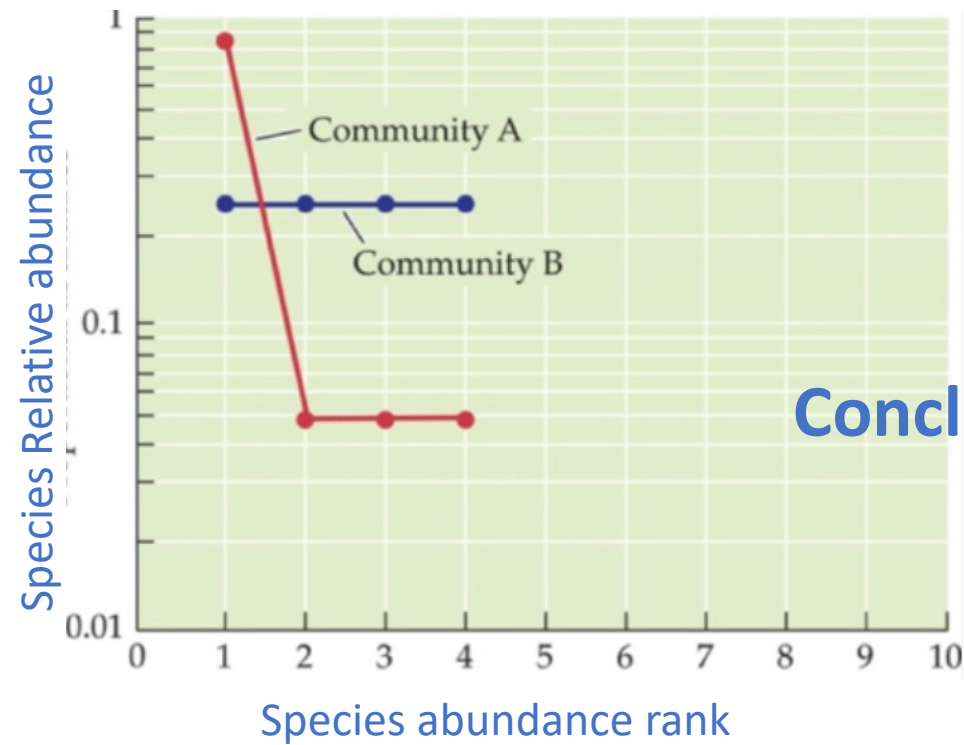


→ Issue : DADA2 remove singletons !!!

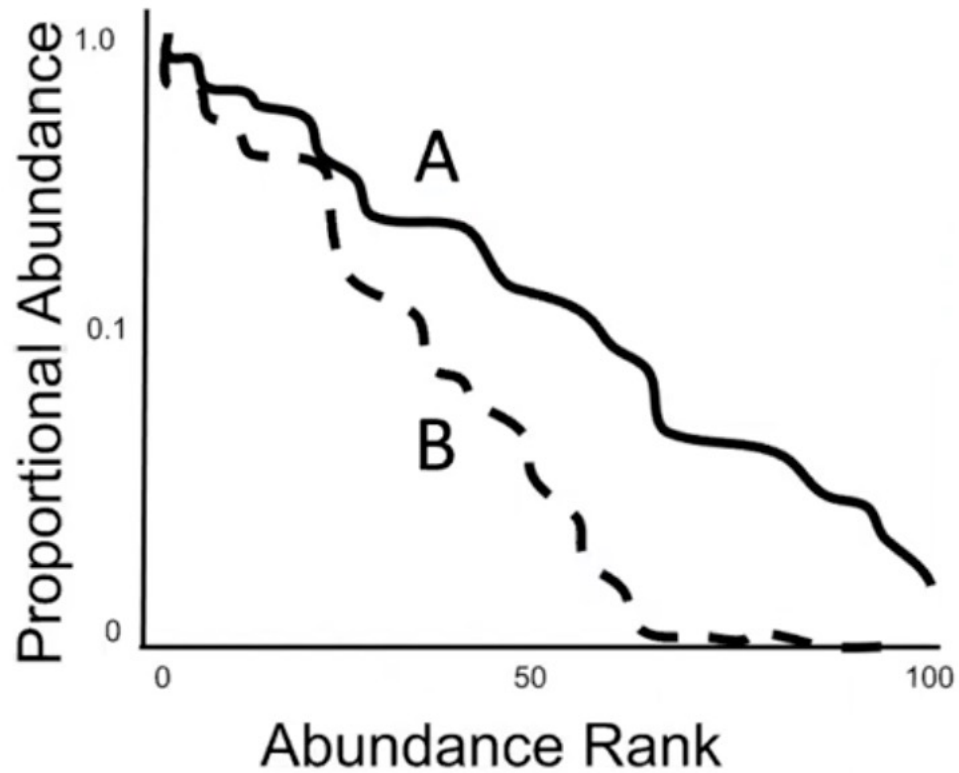
Rank Abundance Curves

Plot relative abundance of species against their rank in abundance

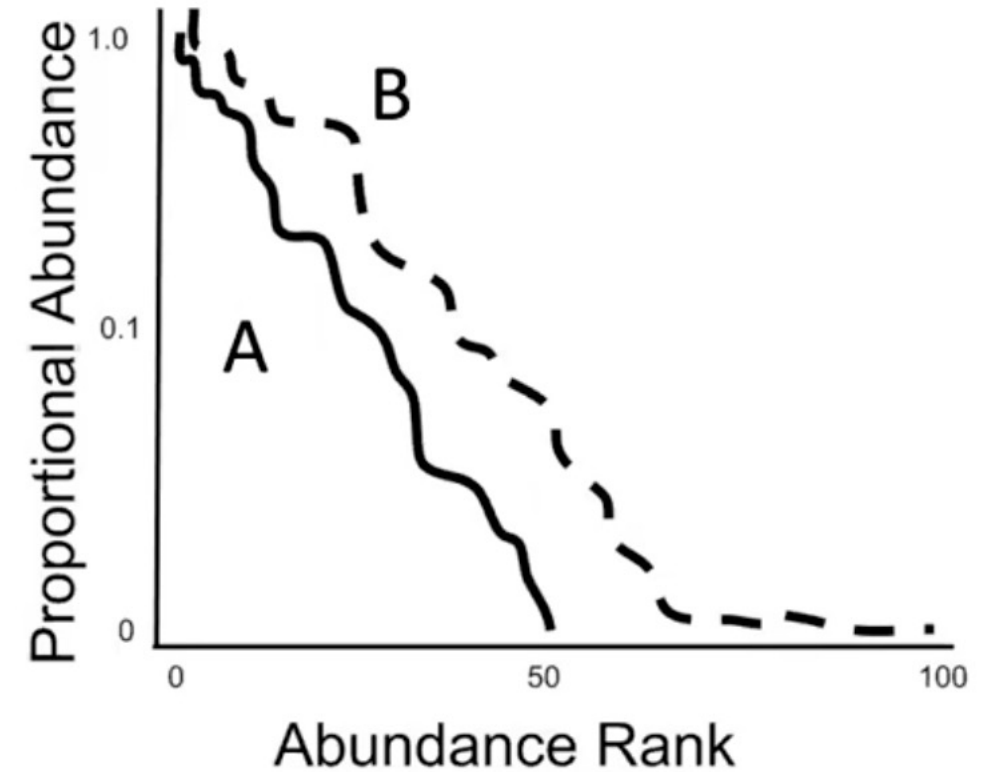
Dominance and diversity... graphically



Conclusions?



- Which one has the greatest evenness?

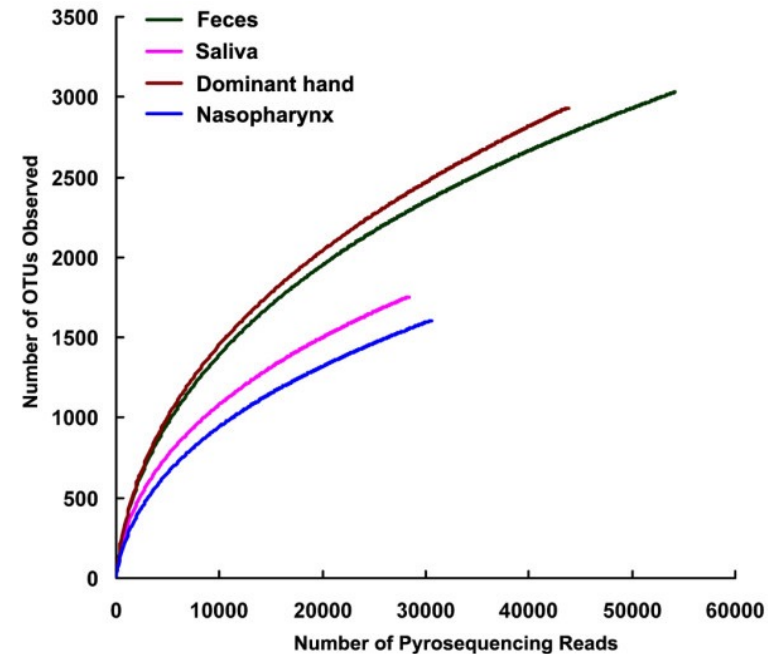
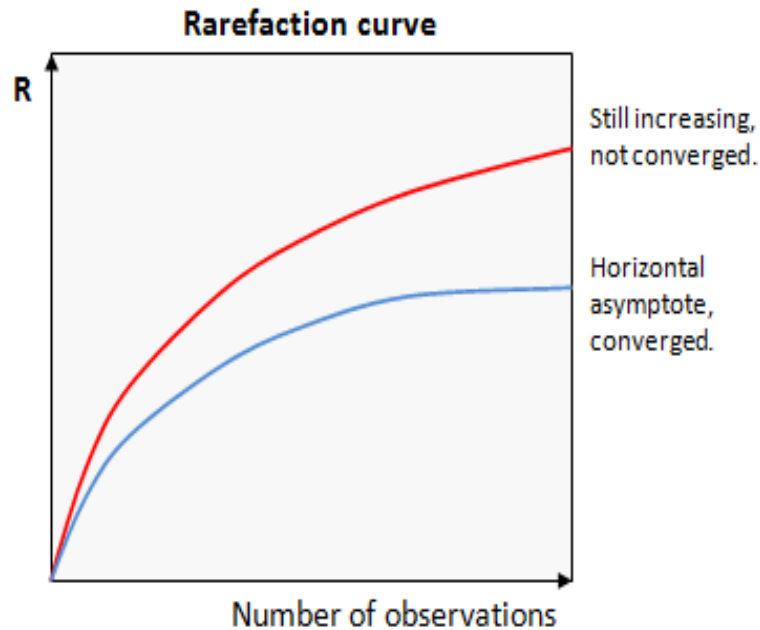


Which one has :

- Greatest evenness?
- Greatest richness?

Rarefaction Curves

« Is the sequencing effort performed (sequencing depth) for a sample (s) sufficient for the number of species observed ? »



→ Reach the asymptote ???

Asymptote means that sequencing more (depth), will not increase your number of OTU/ASVs observed

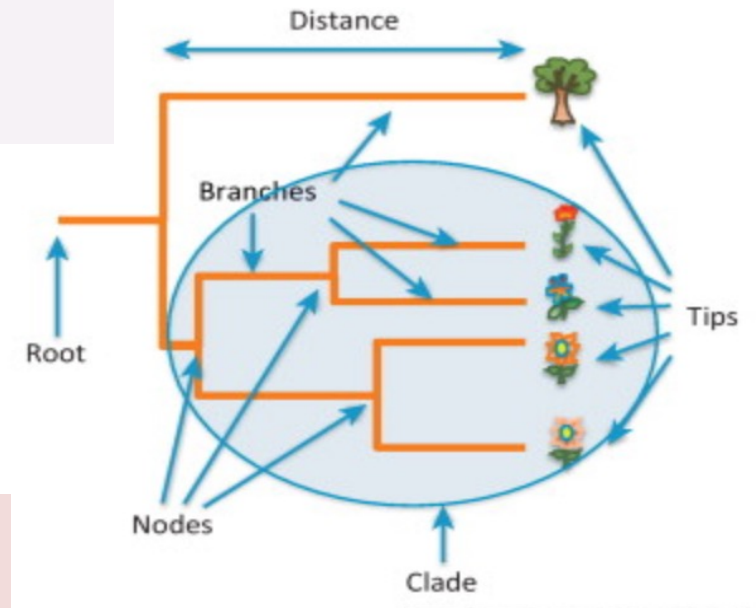
Phylogenetic Indices

Phylogenetic Diversity (PD) measure of the evolutionary history within a set of species :

- Relatedness, speciation, events ...

→ describes a fundamental aspect of biodiversity

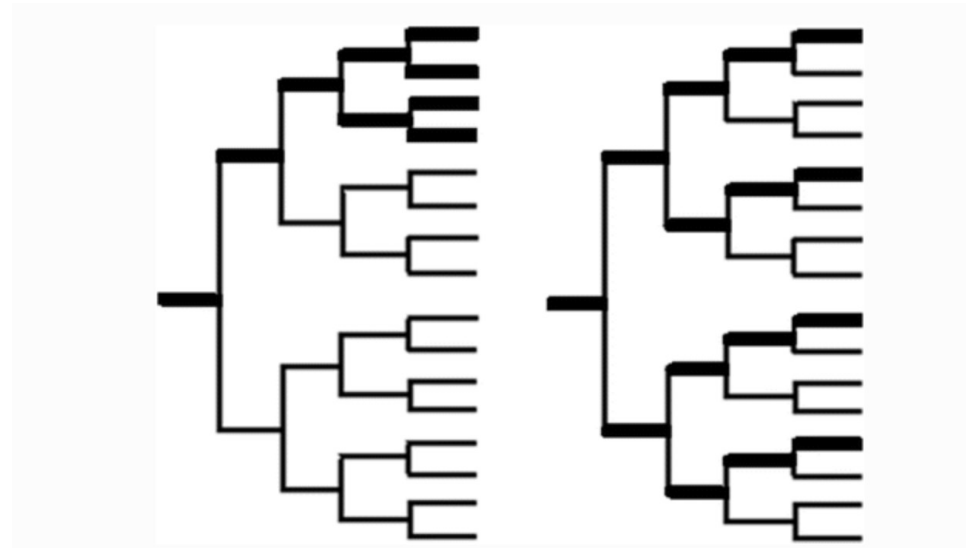
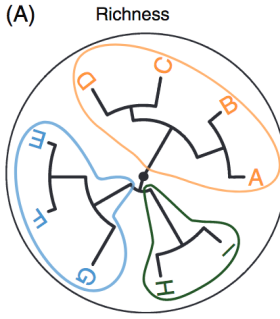
A better predictor of ecosystem function than species richness & evenness



Richness = How much ?

- PD_{faith} (Faith's Phylogenetic Diversity)

→ As the minimal total length of all the branches required to span a given set of taxa to the root (Related to SR)

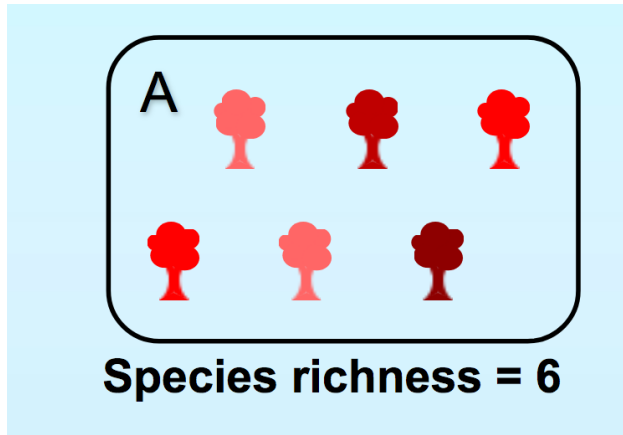


Low PD

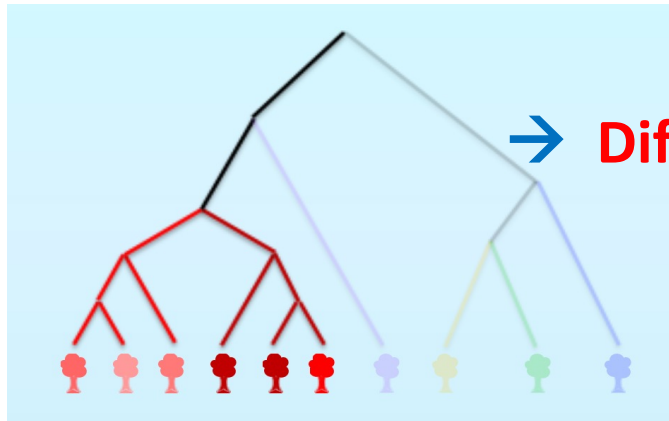
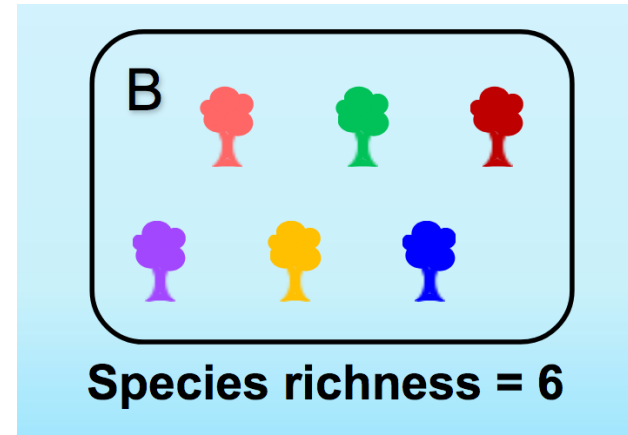
Assemblage of taxa with
more recent evolutionary
history

High PD

Communities with more
evolutionary divergent taxa and
older history

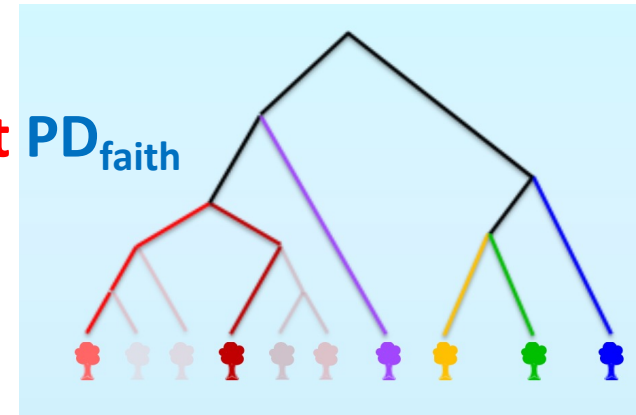


Same SR



EnvA : Sum up branch lengths

→ Different PD_{faith}

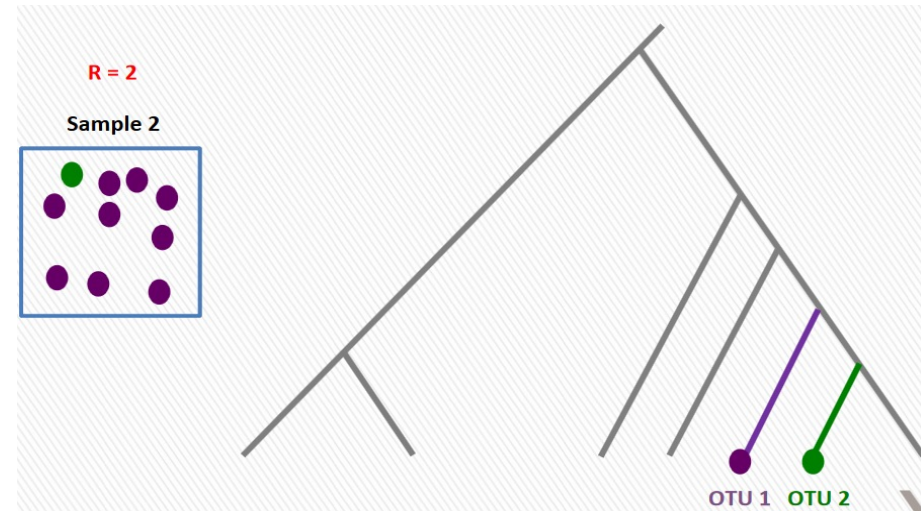
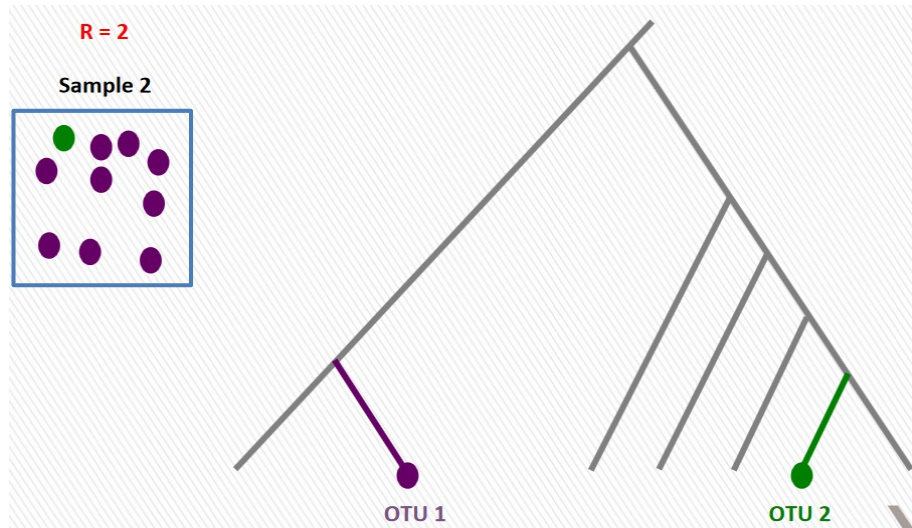


EnvB : Sum up branch lengths

If not using phylogenetic indices : lost of this information !

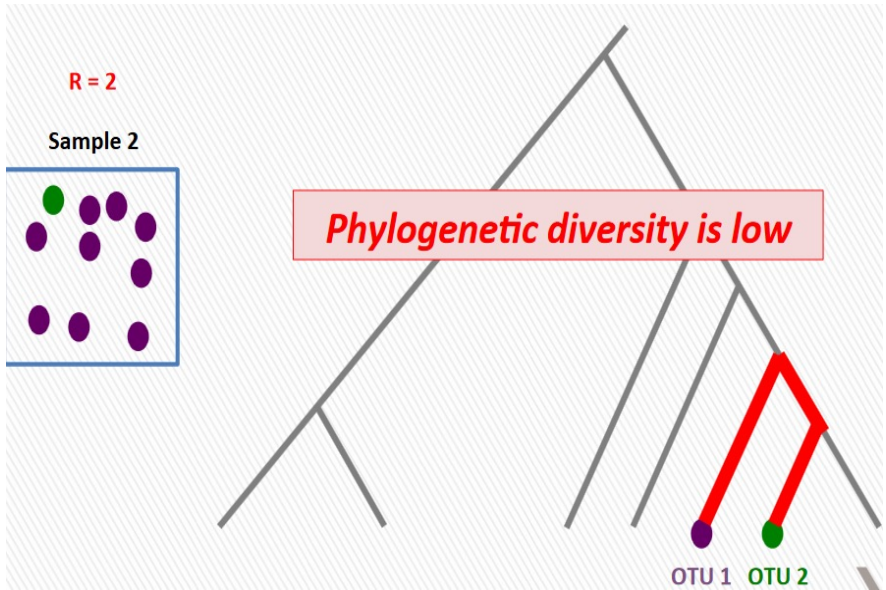
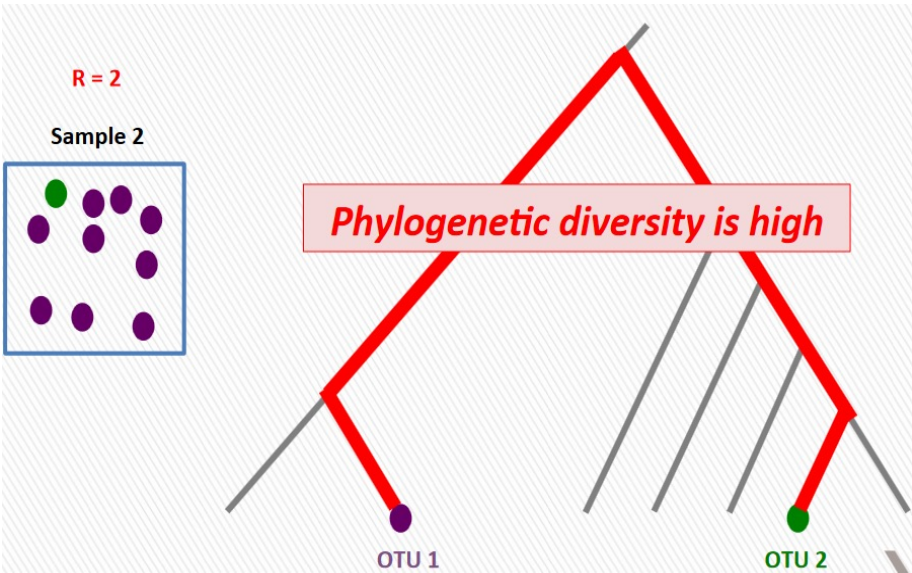
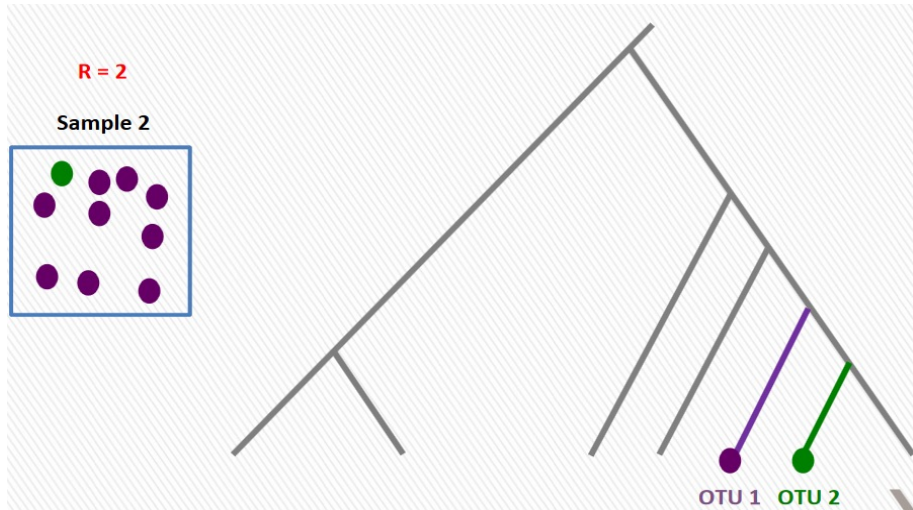
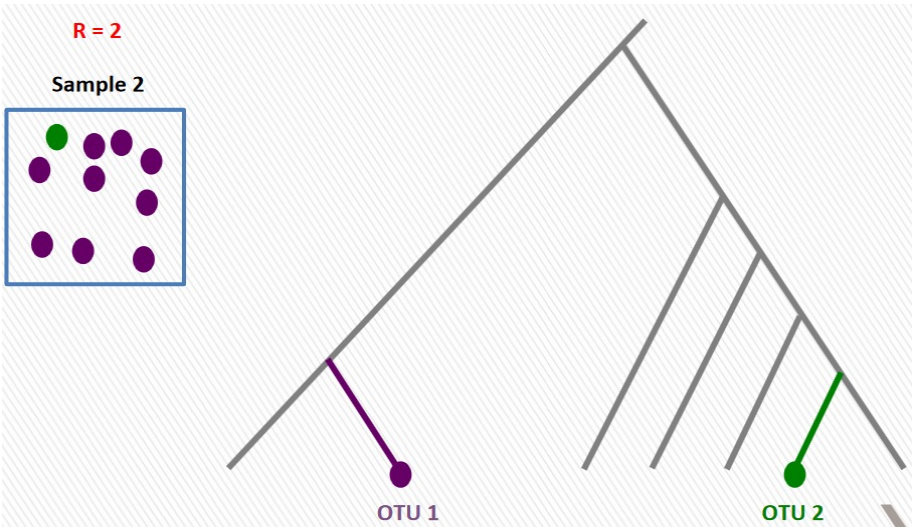
Higher PD

Same SR & same evenness = Same Shannon/simpson



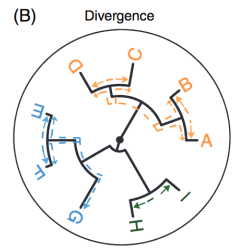
Same phylogenetic diversity ???

Same SR & same evenness = Same Shannon/simpson



BUT different PD_{faith}!!!!

Divergence : Quantify the phylogenetic difference...



Why PD is a proxy of functional diversity, niche/community dissimilarity :

- Closely related species tend to have similar functions/traits (similar habitat requirement)
- Distant related species tend to have greater complementary functions

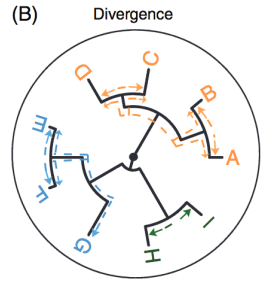
Meaning for the community assemblage :

Dominance of closely related species → **Clustering pattern**

Dominance of distant related species → **Overdispersion pattern**

Given species richness,
does the **phylogenetic diversity** in **AN** assemblage
is greater or less than that expected?

Divergence : How different?



Two commonly used metrics were used to quantify:
the Net Relatedness Index (NRI) and Nearest Taxon Index (NTI)

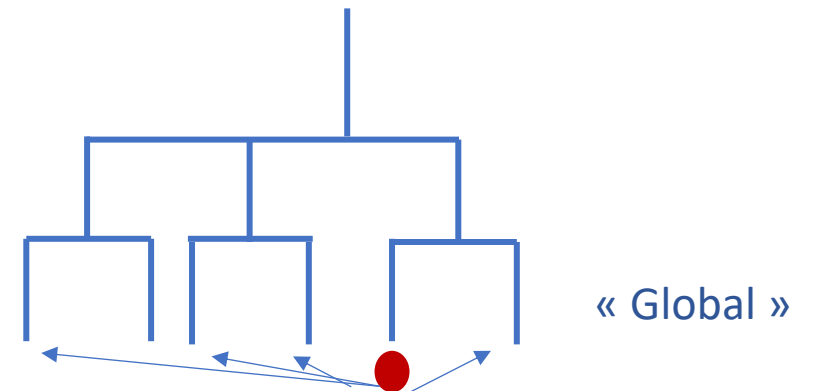
Highlight phylogenetic structure of assemblages **at different evolutionary depths**

NRI : Net Relatedness Index

- Based on the **Mean Phylogenetic Distances (MPD)** in each community.

→ **Average** phylogenetic **distance** of species (**to every other species**)

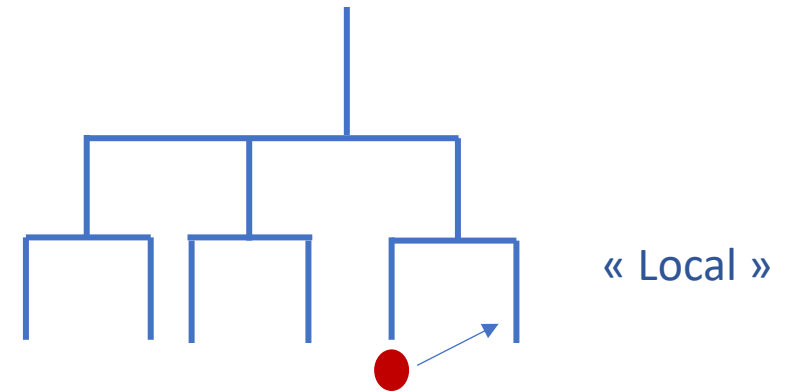
→ « Basal measure » : Clade representation. **Strongly influenced by the 'basal' structure** of the phylogenetic tree



NTI : Nearest Taxonomy Index

- Based on **MNTD** the Mean Nearest phylogenetic Neighbor Distance
→ **Average** phylogenetic **distance** to the **nearest neighbour**

→ Reflect Phylogenetic structure of the tree tips



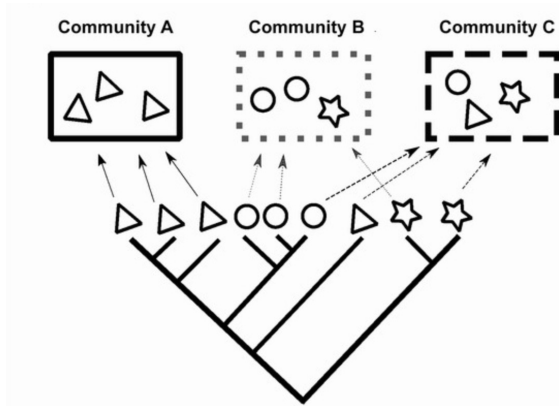
So get NRI /NTI values .. And so what ????

The « Null model » : Phylogeny randomization...

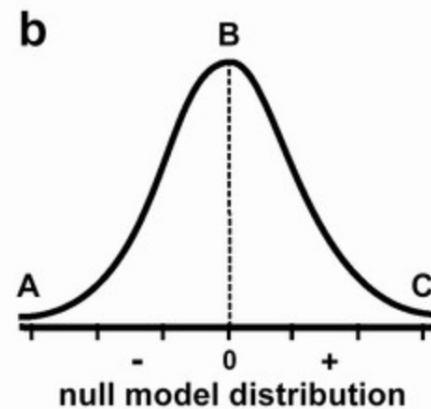
Need a **reference** for comparison → Absence/overdispersion/Clustering!

→ A distribution « Null Model » based on random taxa positions within tree

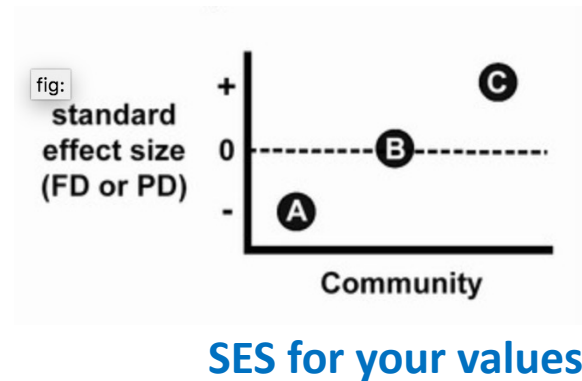
→ Is the measure for a specific community is more or less expected by chance ?



Randomization

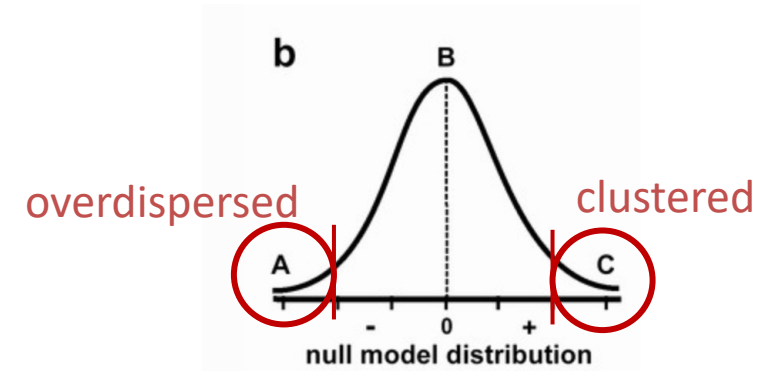


NTI and NRI are Z scores!!



Interpretation

- A negative NRI/NTI value indicates an **overdispersed phylogeny** where taxa are less related to each other than expected by chance
→ Significance < -1.96
- Positive NRI/NTI values indicate a **clustered phylogeny** where taxa are more related to each other than expected by chance
→ Significance > 1.96



Community Assembly : Spatial and Temporal processes

- Niche-driven = Deterministic

- **Selection**

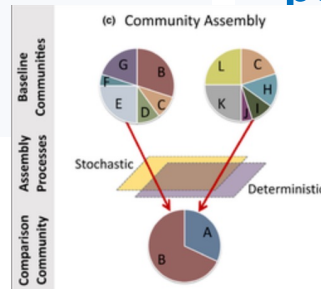
- Biotic interactions (taxa interactions)
- Environmental filtering
(=Abiotic conditions, physico-chemical)

- Neutral = Stochastic process

- **Unpredictable**

- Random proliferation, dispersal
- Random birth-death events

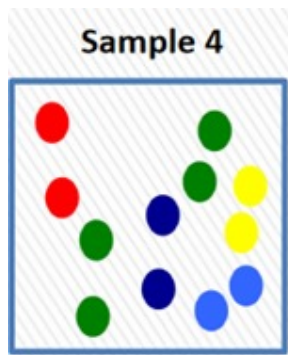
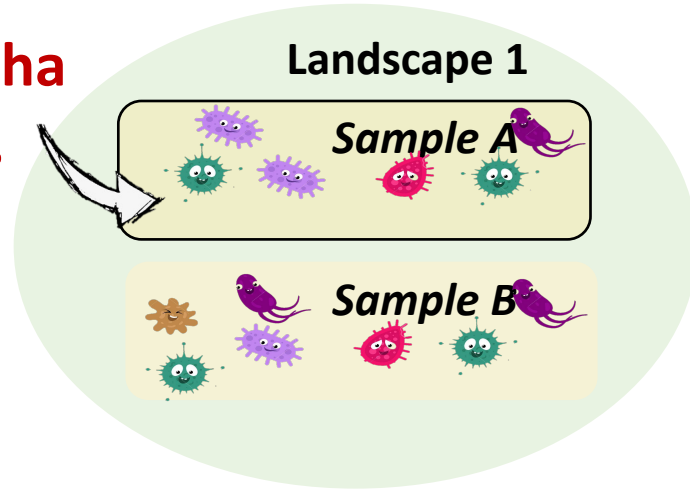
Ecological drift (loss diversity, small pop)



→ See β NTI/NRI

Sum up ...

Alpha
div.

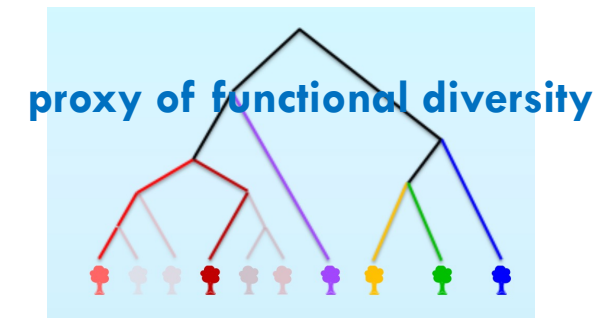


Shannon, Simpson, Pielou ...

Richness & equitability

Neutral diversity indices
(taxonomic)

Phylogenetic indices



NTI/NRI

Be aware of the biases/limitations associated with the indices you use when interpreting your results !!

Diversity within an habitat of fixed size

Use α -diversity indices to quantify the biodiversity of an habitat/sample